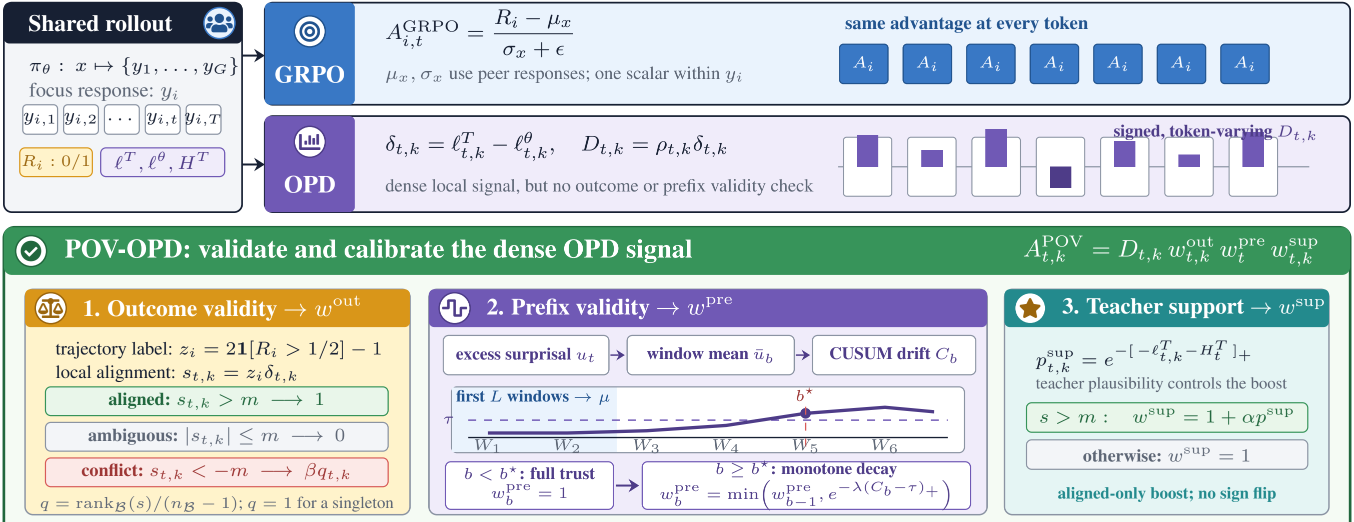

How the three weights change credit on the same response

$b_t \equiv 1 + \alpha p^{\mathrm{sup}}_t$ for aligned tokens; $\gamma_6 \le \gamma_5 < 1$ from drift onward

sampled-token example	t_1 aligned	t_2 aligned	t_3 ambiguous	t_4 conflicting	t_5 aligned + drift	t_6 aligned + later
raw OPD D_t	D_1	D_2	D_3	D_4	D_5	D_6
outcome w^{out}_t	1	1	0	βq_4	1	1
prefix w^{pre}_t	1	1	1	1	γ_5	γ_6
support w^{sup}_t	b_1	b_2	1	1	b_5	b_6
final A^{POV}_t	$D_1 b_1$	$D_2 b_2$	0	$\beta q_4 D_4$	$\gamma_5 D_5 b_5$	$\gamma_6 D_6 b_6$

Outcome calibrates OPD; it is not added as a separate GRPO reward and never flips D_t's sign.